

Quant Finance — Fundamental Terms

A short, crystal-clear reference — GS Quant Hackathon 2026

1. Returns & Risk Basics

Simple return Fractional price change over one period. $r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$. (The definition this hackathon mandates.)

Log return $\ln(P_t/P_{t-1})$. Adds across time; close to simple return for small moves.

Volatility (σ) Standard deviation of returns — how much they scatter. A risk measure. Annualised: $\sigma_{\text{ann}} = \sigma_{\text{daily}}\sqrt{252}$ (252 trading days/yr).

Variance σ^2 . What the hedge metric (HER) actually minimises.

Covariance / Correlation How two assets move together. Correlation $\in [-1, 1]$ is the scale-free version. High correlation = good hedge candidate.

Basis point (bp) 0.01% = 0.0001. The standard unit for yields, spreads, and costs.

2. Risk-Adjusted Performance

Sharpe ratio Return earned per unit of total risk: $\text{Sharpe} = \frac{\mathbb{E}[r] - r_f}{\sigma}$, where r_f is the risk-free rate. Annualised by $\times\sqrt{252}$. Higher is better; > 1 is good, > 2 is excellent.

Sortino ratio Like Sharpe but divides by *downside* deviation only — penalises losses, not upside swings.

Information ratio (IR) Active return over a benchmark per unit of tracking error: $\text{IR} = \frac{\mathbb{E}[r - r_b]}{\text{std}(r - r_b)}$.

Max drawdown Largest peak-to-trough drop in cumulative value. Measures worst-case pain.

Alpha (α) / Beta (β) From $r = \alpha + \beta r_{\text{market}} + \varepsilon$: β is sensitivity to the market; α is excess return unexplained by it. Your proxy “weights” are betas.

R^2 Fraction of variance the model explains (0–1). $R^2 = 1$ is a perfect fit.

3. Fixed Income & This Problem

NAV Net Asset Value — fair per-share value of an ETF’s holdings. **Part 1** estimates it from proxy returns.

Duration Sensitivity of a bond’s price to interest-rate moves (in years).

DV01 Dollar Value of 01 — \$ P&L for a **1bp move in yield**. The **rates** risk number the Part-2 evaluator computes.

CS01 \$ P&L for a **1bp move in credit spread**. The **credit** risk number in Part 2.

Credit spread Extra yield a risky bond pays over a Treasury — compensation for default risk. Widens when markets fear defaults.

Rates vs Credit The two orthogonal risk families of a bond ETF: yield-curve moves (DV01) vs default-spread moves (CS01).

Par / notional Face value of a position. “100 mm notional” = \$100M face. Futures, bonds and CDX are quoted per 100 par; ETFs are quoted per share.

4. Hedging Metrics (Part 3)

Hedge Offsetting position that cancels the risk of another. Static = set once, held.

Hedge Effectiveness Ratio (HER) How much variance the hedge removes: $\text{HER} = 1 - \frac{\text{Var}(\text{Net P\&L})}{\text{Var}(\text{Portfolio P\&L})}$.
HER = 1 = perfect hedge.

Tracking error (TE) Volatility of the leftover P&L. Here: $TE = \frac{\text{StdDev}(\text{Net})}{\text{StdDev}(\text{Portfolio})}$. Lower = tighter hedge.

Hedge ratio How many units of the hedge instrument per unit of exposure (often a β).

Transaction cost What you pay to trade, charged in bp of notional. A one-time penalty at inception here (ETFs 2bp, futures/bonds 0.5bp, CDX 1bp).

5. Error Metrics

MAPE *Mean Absolute Percentage Error*: $\text{mean}\left(\frac{|\hat{y} - y|}{|y|}\right)$. Scale-free; the **Part-1 NAV** score (lower is better).

MAE / RMSE Mean absolute error; root-mean-square error (RMSE punishes big misses harder).

L2 error Euclidean distance $\sqrt{\sum(a - b)^2}$ — how Part 2 compares your DV01/CS01 to the benchmark.

6. Modelling & Regression

OLS *Ordinary Least Squares* — fit coefficients by minimising squared error. Unbiased but overfits with many correlated features.

Regularisation Adding a penalty on coefficient size to curb overfitting.

Ridge (L2) Penalises $\sum w^2$ — shrinks all weights, keeps them all non-zero (dense).

Lasso (L1) Penalises $\sum |w|$ — drives weak weights to *exactly zero* (sparse / feature selection).

ElasticNet Blend of L1+L2 (**l1_ratio**) — sparse *and* stable under collinearity.

Overfitting Model memorises training noise, fails on new data. The enemy of a hidden test set.

Cross-validation (CV) Split data into folds to tune hyper-parameters without peeking at the test set.

TimeSeriesSplit Walk-forward CV: always train on the past, validate on the future — no leakage.

Winsorising Clipping extreme returns to a percentile (e.g. 1%/99%) so outliers don't distort the fit.

Collinearity Features that move together (e.g. TY $\leftrightarrow$ UST_10Y); makes plain OLS weights unstable.

Rule of thumb: returns measure reward, volatility measures risk, and every "ratio" above is just one divided by the other.